

Fibrewise Characterization of Resolution Geometry

Fixed-Protocol Representation, Maximal Admissibility, and a Finite Uniqueness Attack

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Abstract

We formulate Resolution Geometry (RG) for broad observation systems as a modular representation theory of experiment-supported distinctions. The primitive input is a declared experiment together with an admissibility protocol. In the measure-theoretic classical module an experiment is an arbitrary family of probability measures on a measurable outcome space; no domination, finite dimension, density, or smooth parameterization is required for the equality-of-law quotient. Observational extensionality forces the quotient and its universal factorization. Blackwell randomization and Le Cam deficiency then supply exact and approximate comparison layers. Local differential geometry is optional: where regularity exists, a separately stated metric module selects Fisher–Rao geometry in the regular finite classical domain.

For finite-dimensional quantum experiments, preparations are density operators, declared observations are POVMs or quantum-to-classical channels, and equality of all declared output laws defines the operational quotient. A tomographically complete measurement family reduces this equivalence to equality of density operators. Under quantum channels, distinguishability is contractive, but unlike the classical case monotonicity does not select a unique Riemannian metric: the Petz theorem yields a family indexed by symmetric operator-monotone functions. Consequently quantum RG carries a declared quantum metric module, or the complete monotone family, rather than silently choosing SLD/Bures, Bogoliubov–Kubo–Mori, or another member. Quantum sufficiency is represented by recoverability on the declared state family. The same quotient, projector, maximal-gate, naturality, and gluing architecture then applies whenever the chosen module supplies the required positive comparison operator and greatest admissible subobjects.

The fixed-protocol representation theorem states that the resulting object is unique up to protocol-preserving natural isomorphism: every component is either forced by an established universal, comparison, characterization, spectral, greatest-subobject, or cocycle property, or explicitly supplied by the protocol. We also add an operational-time module. A strictly positive self-adjoint coherence semigroup on the maximal observable sector has canonical operator duration $\mathcal{T}_t = -\log C_t = tK$; an extensive continuous conjugation-invariant multiplicative scalar character is a determinant power and yields $a \operatorname{Tr} \mathcal{T}_t$. Compatible clock mechanisms glue to a universal positive affine clock, while identification with relativistic proper time remains conditional on independent causal, rank, path, and held-out calibration evidence. The paper explicitly separates exact theorems, conditional modules, approximate stability statements, and open extensions to channels, instruments, and process tensors. It does not claim protocol-free uniqueness, a unique quantum Fisher metric, latent ontology, or that coherence is physical time.

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1 Introduction

The foundational question is not whether every conceivable observation theory must equal Resolution Geometry. In an unrestricted class, no such uniqueness is plausible: an experiment does not determine a comparison scale, decision threshold, nuisance convention, stability class, or connection. The correct question is conditional:

$$(\mathcal{E}, \Gamma) \longmapsto \text{RG}(\mathcal{E}, \Gamma). \quad (1)$$

Here $\mathcal{E}$ declares the complete observation law and Γ declares the operational contract under which a direction counts as admissibly observable.

This shift is analogous to familiar geometric constructions. A manifold does not select a Riemannian metric, but a Riemannian manifold (M, g) determines its Levi–Civita connection from metric compatibility and vanishing torsion. Likewise, RG does not claim that $\mathcal{E}$ selects Γ . It claims that the bundled input $(\mathcal{E}, \Gamma)$ determines the observable quotient, distinguishability geometry, hard resolution, maximal admissible sector, and compatible transport under explicit hypotheses.

The earlier modular axioms already characterized the quotient, the regular classical metric module, the hard resolved projector, and natural gluing. The remaining finite-dimensional ambiguity was that admissibility had been stated only as containment,

$$W_{\text{obs|adm}} \subseteq \text{im } P_\tau \cap N^\perp \cap S \cap C \cap T, \quad (2)$$

which permits the zero sector and arbitrary strict subspaces. The finite attack reported here isolates the missing condition: admissibility must be complete as well as sound. Thus $W_{\text{obs|adm}}$ is the maximal resolved subobject satisfying all declared gates.

Contribution. The paper provides:

1. an exact definition of the fixed protocol fibre;
2. an admissibility soundness/completeness pair;
3. a proof of maximal-sector uniqueness in finite-dimensional linear fibres;
4. a modular fixed-protocol representation theorem;
5. a natural-isomorphism corollary across transported presentations;
6. an exhaustive finite counterexample search and axiom-independence witness;
7. explicit non-claims separating proved finite results from open extensions.

2 Established mathematical grounding

The result is an assembly theorem. Its load-bearing modules are established constructions used within explicitly stated domains.

2.1 Quotients and universal properties

For a complete statistical experiment $\mathcal{E} = (\Theta, \mathcal{O}, \{P_\theta\}_{\theta \in \Theta})$, equality of laws defines

$$\theta \sim_{\mathcal{E}} \theta' \iff P_\theta = P_{\theta'}. \quad (3)$$

The quotient $Q_{\mathcal{E}} = \Theta / \sim_{\mathcal{E}}$ has the standard factorization property for maps constant on equivalence classes. This is an ordinary quotient universal property; universal constructions are unique up to unique compatible isomorphism. The claim made here is no stronger than that standard fact at the quotient layer.

2.2 Fisher–Rao characterization

In the regular finite classical setting, a Čencov-type theorem characterizes the Fisher information metric, up to a positive scalar normalization, among smooth metric families invariant under sufficient Markov embeddings. Modern extensions establish related formulations for exponential families under explicit invariance hypotheses. RG imports this as a metric module; observational quotient logic alone does not select Fisher geometry.

2.3 Spectral functional calculus

For a positive comparison metric G and positive-semidefinite Fisher tensor F , the information operator

$$L = G^{-1}F \tag{4}$$

is self-adjoint in the G inner product. The finite-dimensional spectral theorem supplies a G -orthonormal eigenbasis, and Borel functional calculus supplies the canonical spectral projection

$$P_\tau = \mathbf{1}_{(\tau, \infty)}(L). \tag{5}$$

The projector is therefore not an extra coordinate after (F, G, τ) is fixed.

2.4 Maximal subobjects

The subspaces of a finite-dimensional vector space form a complete lattice: meets are intersections and joins are linear spans. Consequently, for a finite family of declared linear admissibility subspaces, their intersection is the unique greatest subspace contained in every member. This elementary lattice fact is the established grounding of admissibility completeness in the present theorem. In a more general category, the corresponding construction requires the relevant pullbacks or meets of subobjects; their existence is not assumed automatically outside the finite linear setting.

2.5 Blackwell order and information monotonicity

Blackwell comparison orders statistical experiments by informativeness: a less informative experiment is obtained by garbling a more informative one, equivalently under the classical decision-theoretic hypotheses it cannot improve every decision problem. RG uses only the compatible monotonicity requirement that parameter-independent processing cannot create distinguishability. Blackwell order does not choose G , τ , or the nuisance convention; it constrains admissible processing.

3 Established-theorem timeline and dependency ledger

The following chart is not a novelty claim. It records which established results carry each layer of the construction and prevents the assembled RG theorem from being mistaken for a new proof of its classical modules.

Table 1: Established results underlying the RG assembly.

Date	Result	Established force	Role in RG
1922	Fisher: sufficiency and statistical information	Introduces lossless statistical reduction relative to a model and the information carried by likelihood variation.	Historical basis for experiment-relative information; not itself a quotient-geometry theorem.
1945–1947	Rao–Blackwell theorem	Conditioning an estimator on a sufficient statistic cannot increase convex risk under the usual hypotheses.	Shows that sufficient reductions support essentially complete decision classes; it does not make admissibility select the statistic.
1948	Shannon chain rule	For a deterministic quotient $Q = \pi(X)$, $H(X) = H(Q) + H(X Q)$.	Exact observable/fibre entropy decomposition.
1950	Wald decision theory and complete-class results	Under explicit regularity, admissible procedures are represented within Bayes or generalized-Bayes complete classes.	Decision layer and risk geometry; no protocol-free observational quotient follows.
1951–1953	Blackwell comparison of experiments	Garbling orders experiments by decision informativeness; statistical equivalence preserves decision problems.	Information monotonicity and comparison of observable experiments.
1950s	Minimal sufficiency and Lehmann–Scheffé theory	Minimal sufficient statistics, when they exist in the stated category, are unique up to measurable equivalence; completeness yields uniqueness of unbiased estimators through the statistic.	Canonical statistical reduction under its own assumptions, distinct from rule admissibility.
1955	Basu theorem	A boundedly complete sufficient statistic is independent of every ancillary statistic under the theorem’s hypotheses.	Clarifies informative versus ancillary structure; does not erase latent variables.
1960s	Le Cam comparison and deficiency	Quantifies approximate equivalence and loss between experiments.	Natural route to approximate RG equivalence beyond exact isomorphism; not proved here.
1970s–1982	Čencov characterization	In the regular finite classical category, sufficient-Markov invariance characterizes Fisher–Rao geometry up to scale.	Metric module only; quotient logic alone cannot select Fisher.

Date	Result	Established force	Role in RG
Classical	Spectral theorem and functional calculus	A self-adjoint information operator has canonical spectral projectors.	Uniqueness of P_τ once (F, G, τ) is fixed.
1976	Lindblad–Gorini–Kossakowski–Sudarshan generators	Characterize norm-continuous finite-dimensional quantum dynamical semigroups.	Quantum process/time module; not a general process-tensor theorem.
1990s	Petz monotone-metric classification	Classifies CPTP-monotone quantum Riemannian metrics by operator-monotone functions.	Proves a quantum metric family, not uniqueness of SLD/Bures.
2000s–present	Quantum combs and process tensors	Encode multitime intervention statistics and operational memory.	Proper domain for quantum memory beyond BCH commutators.
Classical	Subspace lattices and categorical uniqueness	Intersections are greatest common subspaces; universal/greatest objects are unique by structure-preserving isomorphism.	Maximal admissible sector and fibrewise canonicity.

3.1 The essential correction to the decision-theory reading

The strongest defensible statement is not “admissibility forces a unique sufficient statistic.” Admissibility is defined only after an action space and loss are fixed, whereas sufficiency and minimal sufficiency are properties of the statistical experiment. A sufficient statistic permits Rao–Blackwell improvement for convex losses, and complete-class theorems relate admissible rules to Bayes procedures under regularity. Neither implication reverses into protocol-free selection of a unique observation geometry. RG closes this logical gap by declaring Γ and then imposing both admissibility soundness and admissibility completeness. The quotient is universal; the final sector is greatest only inside its fixed protocol fibre.

Proposition 3.1 (Separation of statistical and protocol uniqueness). *Let $\mathcal{E}$ be a statistical experiment and let Γ be a fixed admissibility protocol. Minimal sufficiency, when available, characterizes a reduction of $\mathcal{E}$ up to the relevant measurable equivalence. The RG sector $W_{\text{obs|adm}}$ is instead characterized as the greatest subobject satisfying the gates of Γ . Decision-rule admissibility alone implies neither construction.*

Proof. Minimal sufficiency is ordered among sufficient statistics and contains no loss function or threshold. Decision admissibility is ordered by risk and therefore depends on the declared decision problem. The RG sector additionally depends on G , τ , nuisance, stability, conditioning, coarse-graining, and transport data. Since these inputs can vary while $\mathcal{E}$ remains fixed, no implication from decision admissibility alone can determine them. Once Γ is fixed, the greatest-subobject argument below gives uniqueness of $W_{\text{obs|adm}}$. $\square$

4 Declared admissible experiments

Definition 4.1 (Complete statistical experiment). *A complete statistical experiment is*

$$\mathcal{E} = (\Theta, \mathcal{O}, \{P_\theta\}_{\theta \in \Theta}), \quad (6)$$

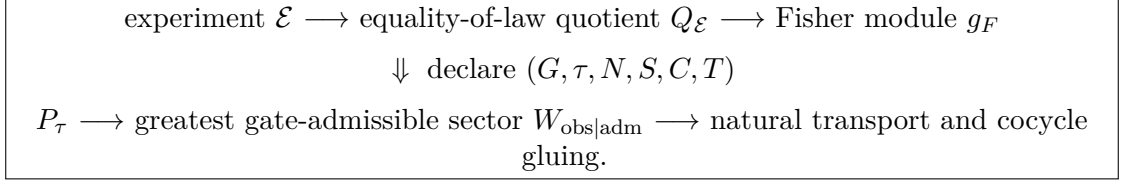


Figure 1: Logical dependency of the final RG object. Vertical declaration marks the point at which protocol-relative structure enters.

where Θ is a parameter or preparation domain, $\mathcal{O}$ is an outcome space, and P_{θ} is the complete outcome law at θ .

Definition 4.2 (Admissibility protocol). *An admissibility protocol is a tuple*

$$\Gamma = (G, \tau, N, S, C, T), \quad (7)$$

where G is a covariant positive comparison structure, τ is a resolution threshold or threshold family, N is a nuisance/gauge convention, S is a stability protocol, C is a conditioning and coarse-graining protocol, and T is transport/gluing data.

Definition 4.3 (Declared admissible experiment). *The primitive RG object is the bundled pair*

$$\mathbb{E} = (\mathcal{E}, \Gamma). \quad (8)$$

No component of Γ is inferred silently from inaccessible latent labels, undeclared priors, or an external physical interpretation.

Definition 4.4 (Protocol fibre). *Fix a protocol Γ . The protocol fibre Exp_{Γ} is the category whose objects are experiments carrying that protocol type and whose morphisms are experiment morphisms together with the prescribed covariant transport of all protocol data. The target fibre ObsGeom_{Γ} consists of quotient geometries, metric modules, resolved projectors, admissible sectors, and transition maps preserving the fixed contract.*

Remark 4.5. “Fixed protocol” means fixed as structured data, not fixed coordinate arrays. Under a change of representation, G , nuisance data, gates, and transports move covariantly. Fibrewise uniqueness is therefore uniqueness up to protocol-preserving natural isomorphism, not literal equality of matrices.

5 Broad statistical experiments and comparison geometry

5.1 Measure-theoretic experiment core

Definition 5.1 (General statistical experiment). *A classical statistical experiment is a triple*

$$\mathcal{E} = (\Theta, (\Omega, \mathcal{A}), \{P_{\theta} : \theta \in \Theta\}),$$

where $(\Omega, \mathcal{A})$ is an arbitrary measurable space and each P_{θ} is a probability measure. No common dominating measure, topology on Θ , finite-dimensional parameterization, density, or smoothness is part of this definition.

Theorem 5.2 (General quotient representation). *For every general statistical experiment, equality of laws*

$$\theta \sim_{\mathcal{E}} \theta' \iff P_{\theta} = P_{\theta'}$$

is an equivalence relation. Every extensional map $\Phi : \Theta \rightarrow Z$ factors uniquely through $\Theta / \sim_{\mathcal{E}}$.

Proof. Equality of measures is reflexive, symmetric, and transitive. Define $\tilde{\Phi}([\theta]) = \Phi(\theta)$. Extensionality gives representative independence and surjectivity of the quotient map gives uniqueness. No domination or regularity is used. $\square$

Definition 5.3 (Randomization and Blackwell preorder). *For experiments $\mathcal{E} = (P_\theta)$ and $\mathcal{F} = (Q_\theta)$ with the same parameter set, write $\mathcal{E} \succeq_B \mathcal{F}$ when a parameter-independent Markov kernel K satisfies $Q_\theta = KP_\theta$ for every θ . Mutual randomizability is Blackwell equivalence.*

Proposition 5.4 (Functorial information order). *Any RG distinguishability functional declared monotone under Markov kernels is nonincreasing along $\mathcal{E} \succeq_B \mathcal{F}$. Blackwell-equivalent experiments therefore receive isomorphic comparison data whenever the construction is natural with respect to the witnessing kernels.*

Proof. The first statement is the monotonicity axiom applied to K . Apply it in both directions under mutual randomizability. Naturality identifies the transported structures. $\square$

Definition 5.5 (Le Cam deficiency layer). *On experiment classes where total-variation randomization is available, define*

$$\delta(\mathcal{E}, \mathcal{F}) = \inf_K \sup_{\theta \in \Theta} \|KP_\theta - Q_\theta\|_{\text{TV}}, \quad \Delta(\mathcal{E}, \mathcal{F}) = \max\{\delta(\mathcal{E}, \mathcal{F}), \delta(\mathcal{F}, \mathcal{E})\}.$$

Normalization conventions for total variation may insert a factor 1/2.

Proposition 5.6 (Approximate transfer is an additional module). *If an RG output functional A obeys a declared modulus of continuity*

$$d(A(\mathcal{E}), A(\mathcal{F})) \leq \omega(\Delta(\mathcal{E}, \mathcal{F})), \quad \omega(r) \rightarrow 0 \text{ as } r \rightarrow 0,$$

then asymptotically equivalent experiments transfer that output. Quotient universality alone does not imply such a bound.

Proof. The conclusion is immediate from the modulus. The second statement follows because exact factorization only identifies equal laws and contains no quantitative topology on experiments or outputs. $\square$

5.2 Stratified hierarchy rather than a universal metric

The broad theory has four nested layers:

1. the exact equality-of-law quotient, valid for every experiment;
2. exact or approximate comparison by randomizations and deficiency;
3. tangent or differentiable structures only on regular strata;
4. metric and spectral sectors only after a metric module and protocol are declared.

For non-dominated, singular, infinite-dimensional, or rank-changing experiments, Fisher matrices may fail to exist, be infinite, or change rank. RG therefore retains the quotient/comparison layers and treats local geometry chartwise on declared regular strata. This is an extension of scope, not an assertion that every statistical experiment is a smooth manifold.

Non-claim 5.7 (No general complete-class shortcut). *The existence of an admissible decision rule does not imply existence of a finite-dimensional sufficient statistic, a complete statistic, or a unique RG protocol. Complete-class, Bayes, and Rao–Blackwell results retain their own loss, compactness, convexity, measurability, and regularity hypotheses.*

6 Quantum statistical experiment module

6.1 Preparations, measurements, and operational quotient

Definition 6.1 (Finite-dimensional quantum experiment). *A quantum preparation experiment is*

$$\mathcal{E}_q = (\Theta, \mathcal{H}, \{\rho_\theta\}_{\theta \in \Theta}, \mathfrak{M}),$$

where ρ_θ are density operators on a finite-dimensional Hilbert space and $\mathfrak{M}$ is the declared family of POVMs, instruments, or quantum-to-classical channels. For a POVM $M = \{M_x\}$ the outcome law is $p_\theta^M(x) = \text{Tr}(\rho_\theta M_x)$.

Definition 6.2 (Quantum observational equivalence). *Define*

$$\theta \sim_{\mathcal{E}_q} \theta' \iff p_\theta^M = p_{\theta'}^M \text{ for every } M \in \mathfrak{M}.$$

The quotient $Q_{\mathcal{E}_q} = \Theta / \sim_{\mathcal{E}_q}$ is relative to the declared measurement family.

Theorem 6.3 (Quantum quotient factorization). *Every construction determined solely by all declared quantum outcome laws factors uniquely through $Q_{\mathcal{E}_q}$.*

Proof. The family of outcome laws defines an ordinary equality relation, so the general quotient representation theorem applies. $\square$

Proposition 6.4 (Tomographic closure). *If $\mathfrak{M}$ separates density operators, then*

$$\theta \sim_{\mathcal{E}_q} \theta' \iff \rho_\theta = \rho_{\theta'}.$$

If $\mathfrak{M}$ is not separating, distinct density operators may remain in one observational class.

Proof. The forward implication is the separating property; the reverse follows from the Born rule. Failure of separation is precisely existence of distinct states with equal declared outcomes. $\square$

6.2 Quantum channels, sufficiency, and recoverability

Definition 6.5 (Quantum randomization preorder). *For state families on possibly different Hilbert spaces, write $\mathcal{E}_q \succeq_Q \mathcal{F}_q$ if a completely positive trace-preserving map Φ satisfies $\sigma_\theta = \Phi(\rho_\theta)$ for all θ .*

Definition 6.6 (Sufficiency by recovery). *A channel Φ is sufficient for the declared family $\{\rho_\theta\}$ when there exists a CPTP recovery channel $\mathcal{R}$ such that*

$$(\mathcal{R} \circ \Phi)(\rho_\theta) = \rho_\theta \quad \forall \theta.$$

This is the quantum analogue of lossless randomization on the specified family; it is not invertibility on all operators.

Proposition 6.7 (Recovery equivalence). *A sufficient channel and its recovery witness make the input and output state families mutually quantum-randomizable. Every natural monotone RG module must agree on them up to the induced isomorphism.*

Proof. The forward channel gives one randomization and the recovery map gives the reverse on the family. Apply monotonicity and naturality in both directions. $\square$

6.3 Petz monotone metric family

Assumption 6.8 (Faithful regular quantum stratum). *On a smooth family of faithful density operators, tangent vectors are traceless self-adjoint operators and admissible coarse-grainings are CPTP maps.*

Theorem 6.9 (Quantum monotone-metric module). *On finite-dimensional faithful state manifolds, quantum monotone Riemannian metrics are classified by symmetric normalized operator-monotone functions f . One convenient form is*

$$g_\rho^{(f)}(A, B) = \text{Tr}[A c_f(L_\rho, R_\rho)(B)], \quad c_f(x, y) = \frac{1}{y f(x/y)},$$

where $L_\rho(X) = \rho X$ and $R_\rho(X) = X \rho$. Each member contracts under CPTP maps on its domain.

Grounding. This is the Petz classification theorem imported as the quantum metric module. RG does not reprove it. $\square$

Corollary 6.10 (Classical restriction). *On commuting state families, the monotone quantum metrics restrict, with the common normalization convention, to the classical Fisher–Rao metric.*

Non-claim 6.11 (No unique quantum Fisher metric). *CPTP monotonicity does not select one quantum metric. SLD/Bures, Bogoliubov–Kubo–Mori, Wigner–Yanase-type, and other Petz metrics encode different operational choices. A quantum RG object must therefore include f or another metric-selection principle in its protocol, or retain the complete monotone family.*

Definition 6.12 (Quantum RG fibre). *For a declared quantum protocol Γ_q containing a measurement family, monotone metric label or family, comparison structure, threshold, nuisance convention, stability rules, admissible operations, and transport data, define*

$$\text{RG}_q(\mathcal{E}_q, \Gamma_q) = (Q_{\mathcal{E}_q}, \{g^{(f)}\}_{f \in \mathcal{F}_q}, P_{\tau, f}, W_{\text{obs|adm}}^{(f)}, \{T_{ij}\}, \Gamma_q),$$

whenever the spectral and greatest-subobject hypotheses hold.

Theorem 6.13 (Conditional quantum fibrewise representation). *Fix a finite-dimensional quantum experiment and a complete protocol Γ_q . Assume: the declared measurement laws are complete for the intended experiment; a faithful regular stratum or support convention is fixed; a Petz metric member or covariant family is declared; the comparison operator is positive with threshold away from its spectrum; the quantum gate family admits a greatest admissible resolved subobject; and transports satisfy naturality, identity, and cocycle laws. Then $\text{RG}_q(\mathcal{E}_q, \Gamma_q)$ is unique up to protocol-preserving natural isomorphism and the declared metric normalization.*

Proof. Quantum outcome laws give the universal quotient. The protocol fixes the Petz metric member or family. Spectral functional calculus fixes each hard projector. Soundness and completeness fix the greatest admissible sector. Naturality and cocycle laws fix transported copies and gluing. These are the same categorical uniqueness steps as in the classical fibre, with the metric module replaced. $\square$

Remark 6.14 (Channels, instruments, and process experiments). *A channel experiment may be tested by preparation-ancilla-measurement strategies; a process experiment may be tested by multitime instruments. In each case the safest quotient is equality of all declared tester outcome laws. Equality of Choi operators is available only when the tester class is tomographically complete and finite-dimensional assumptions hold. The quotient theorem survives unchanged; metric, memory, and composition modules require separate characterizations.*

7 Axioms for fixed-protocol observable geometry

Axiom 7.1 (Observational extensionality). *Every experiment-internal construction Φ satisfies*

$$P_\theta = P_{\theta'} \implies \Phi(\theta) = \Phi(\theta'). \quad (9)$$

Axiom 7.2 (Naturality). *Observable constructions commute with experiment isomorphisms and with covariant transport of the complete protocol.*

Axiom 7.3 (Information order). *A parameter-independent garbling cannot increase the local distinguishability tensor.*

Axiom 7.4 (Off-discriminant regularity). *Continuous structures vary continuously and discrete invariants remain locally constant away from the declared discriminant, including threshold crossings, rank changes, nuisance-rank changes, conditioning boundaries, stability failures, and transport singularities.*

Axiom 7.5 (Composition and gluing). *Local comparison maps preserve identities and composition. On overlaps,*

$$T_{ii} = \text{id}, \quad T_{jk}T_{ij} = T_{ik}. \quad (10)$$

Assumption 7.6 (Regular classical metric module). *On regular finite classical statistical models, local distinguishability is represented by a continuous smooth tensor family invariant under sufficient Markov embeddings, with fixed positive normalization.*

Axiom 7.7 (Hard spectral resolution). *The resolved selector P is G -self-adjoint, idempotent, commutes with $L = G^{-1}F$, agrees with the threshold mask on every generalized eigenspace, and composes on independent direct sums.*

Definition 7.8 (Gate-admissible resolved subobjects). *Let $R_\tau = \text{im } P_\tau$. For fixed Γ , let $\text{Adm}_\Gamma(R_\tau)$ be the set of resolved subobjects satisfying all declared nuisance, stability, conditioning, coarse-graining, and transport gates.*

Axiom 7.9 (Admissibility soundness). *The reported sector belongs to the admissible family:*

$$W_{\text{obs|adm}} \in \text{Adm}_\Gamma(R_\tau). \quad (11)$$

Equivalently, it contains no direction rejected by any declared gate.

Axiom 7.10 (Admissibility completeness). *The reported sector is greatest among gate-admissible resolved subobjects:*

$$U \in \text{Adm}_\Gamma(R_\tau) \implies U \subseteq W_{\text{obs|adm}}. \quad (12)$$

Thus $W_{\text{obs|adm}}$ retains every direction accepted by the complete declared protocol.

Remark 7.11. *Soundness is a safety condition; completeness is a non-arbitrariness condition. Soundness alone permits arbitrary strict reductions. Completeness forbids discarding an accepted direction without adding a new declared gate.*

8 Characterization ladder

Theorem 8.1 (Quotient representation). *Under observational extensionality, every experiment-internal map $\Phi : \Theta \rightarrow Z$ factors uniquely through $Q_\mathcal{E}$. There is a unique $\tilde{\Phi} : Q_\mathcal{E} \rightarrow Z$ such that*

$$\Phi = \tilde{\Phi} \circ \pi_\mathcal{E}. \quad (13)$$

Proof. Define $\tilde{\Phi}([\theta]) = \Phi(\theta)$. Extensionality makes the definition representative-independent. Surjectivity of $\pi_{\mathcal{E}}$ gives uniqueness. $\square$

Corollary 8.2 (Canonical observable domain). *Any two realizations satisfying the quotient universal property are uniquely isomorphic by an isomorphism commuting with the quotient maps.*

Proof. This is the standard uniqueness argument for universal quotients: each candidate factors uniquely through the other, and uniqueness forces the composites to be identities. $\square$

Theorem 8.3 (Classical metric module). *Under the regular classical metric assumptions, the local distinguishability metric is Fisher–Rao up to the fixed normalization.*

Proof. This is the imported Čencov characterization module. The present paper does not reprove it and uses it only in its stated domain. $\square$

Theorem 8.4 (Unique hard resolved projector). *Let $F \succeq 0$, $G \succ 0$, and $\tau \notin \text{spec}(G^{-1}F)$. The unique operator satisfying hard spectral resolution is*

$$P_{\tau} = \mathbf{1}_{(\tau, \infty)}(G^{-1}F). \quad (14)$$

Proof. The operator $L = G^{-1}F$ is G -self-adjoint and therefore admits a G -orthonormal spectral decomposition. Commutation makes each eigenspace invariant. Idempotence and G -self-adjointness restrict the selector on each eigenspace to an orthogonal projection. Threshold consistency fixes that projection to identity above τ and zero at or below τ . Functional calculus gives the displayed projector and direct-sum composition. $\square$

Lemma 8.5 (Intersection realization of a finite linear protocol). *Suppose the fixed gates of Γ are represented by linear subspaces $A_1, \dots, A_k \subseteq R_{\tau}$. Then*

$$W_* = \bigcap_{a=1}^k A_a \quad (15)$$

is the unique greatest subspace satisfying every gate.

Proof. The intersection is a subspace contained in every A_a , hence it is sound. If U is contained in every A_a , then by the definition of intersection $U \subseteq W_*$. Hence W_* is greatest. If W' is another greatest element, then $W_* \subseteq W'$ and $W' \subseteq W_*$, so $W' = W_*$. $\square$

Theorem 8.6 (Maximal admissible-sector characterization). *In a finite-dimensional linear protocol fibre whose declared gates admit a greatest common resolved subobject, soundness and completeness characterize a unique final sector:*

$$W_{\text{obs|adm}} = \max \text{Adm}_{\Gamma}(R_{\tau}). \quad (16)$$

For linear subspace gates this becomes

$$W_{\text{obs|adm}} = R_{\tau} \cap A_N \cap A_S \cap A_C \cap A_T. \quad (17)$$

Proof. Existence follows from the assumed greatest admissible subobject, or from the intersection lemma in the linear case. If W and W' both satisfy soundness and completeness, completeness of W applied to W' gives $W' \subseteq W$, while completeness of W' applied to W gives $W \subseteq W'$. Thus $W = W'$. $\square$

Corollary 8.7 (No undeclared shrinkage). *Any strict subobject $U \subsetneq W_{\text{obs|adm}}$ violates admissibility completeness unless an additional gate excluding $W_{\text{obs|adm}} \setminus U$ is added to Γ .*

Corollary 8.8 (Conditional uniqueness of $W_{\text{obs|adm}}$). *The final admissible sector is unique conditional on the complete protocol but is not universal with respect to changes of protocol.*

Theorem 8.9 (Natural transport of fixed-protocol geometry). *Let $f : (\mathcal{E}, \Gamma) \rightarrow (\mathcal{E}', \Gamma')$ be an experiment isomorphism transporting the complete protocol. Then f induces an isomorphism of quotient domains, transports the Fisher tensor covariantly, conjugates the hard projector, and maps the maximal admissible sector onto the maximal admissible sector:*

$$f_* W_{\text{obs|adm}} = W'_{\text{obs|adm}}. \quad (18)$$

Proof. Quotient naturality follows from equality-of-law preservation. Metric covariance is part of the metric module. Spectral functional calculus is equivariant under conjugation. Since the complete gate family is transported, $f_* W_{\text{obs|adm}}$ is admissible in the target. For any target-admissible U' , its inverse image is source-admissible and hence lies in $W_{\text{obs|adm}}$; therefore $U' \subseteq f_* W_{\text{obs|adm}}$. Thus the image is target-maximal and equals the unique target sector. $\square$

Theorem 8.10 (Fixed-protocol observable-geometry representation). *Let $(\mathcal{E}, \Gamma)$ be a regular finite-dimensional classical declared admissible experiment. Assume:*

1. *observational extensionality and quotient universality;*
2. *the regular classical Fisher metric module with declared normalization;*
3. *a positive comparison metric G , a threshold τ separated from the generalized spectrum, and the hard spectral-resolution axioms;*
4. *a gate family admitting a greatest admissible resolved subobject;*
5. *admissibility soundness and completeness;*
6. *natural transport, identity, cocycle gluing, and independent-product composition.*

Then the observable geometry

$$\text{RG}(\mathcal{E}, \Gamma) = (Q_{\mathcal{E}}, g_F, P_{\tau}, W_{\text{obs|adm}}, \{T_{ij}\}, \Gamma) \quad (19)$$

is determined uniquely up to protocol-preserving natural isomorphism and the declared Fisher normalization.

Proof. The quotient representation theorem uniquely fixes $Q_{\mathcal{E}}$ up to unique compatible isomorphism. The imported metric characterization fixes g_F up to its declared normalization. The spectral projector theorem uniquely fixes P_{τ} . The maximal admissible-sector theorem uniquely fixes $W_{\text{obs|adm}}$. Naturality uniquely determines transported copies from these fixed local data, while identity and cocycle compatibility make direct and composite identifications agree on overlaps. Product composition excludes a distinct assignment on independent sums. Therefore two realizations of the same bundled input are connected componentwise by protocol-preserving isomorphisms. The quotient map and transported structural maps determine the compatible isomorphism; the componentwise naturality squares commute. Hence the realizations are naturally isomorphic. $\square$

Remark 8.11 (Exact force of established literature). *The quotient and metric layers inherit genuine established characterization theorems. The projector layer is a direct consequence of the spectral theorem and the fixed threshold mask. The admissible-sector layer is a greatest-subobject argument in a subspace lattice. Category theory then supplies the correct uniqueness language: universal or greatest constructions are unique only through structure-preserving maps. The full RG theorem is therefore modular; no cited classical theorem, by itself, states the assembled RG result.*

Corollary 8.12 (Fibrewise canonicity). *For each fixed admissibility protocol Γ , the RG assignment defines a canonical functor*

$$\text{RG}_\Gamma : \text{Exp}_\Gamma \longrightarrow \text{ObsGeom}_\Gamma \quad (20)$$

within the theorem's domain. There is no asserted canonical identification between fibres belonging to different protocols.

Corollary 8.13 (No canonical protocol-free section). *The representation theorem does not provide a map $\mathcal{E} \mapsto \Gamma$. Consequently, it does not define a canonical section of the projection from declared experiments to bare experiments.*

9 Entropy as a quotient corollary

Entropy is not an axiom of RG. It is a downstream construction once a probability measure is placed on a microstate space and pushed through an observational quotient.

Theorem 9.1 (Observable-Quotient Entropy Decomposition). *Let X be a discrete microstate variable and $Q = \pi(X)$ a deterministic observational quotient. Then*

$$H(X) = H(Q) + H(X \mid Q). \quad (21)$$

If $S_G = k_B H(X)$, then

$$\boxed{S_G = k_B H(Q) + k_B H(X \mid Q)}. \quad (22)$$

Proof. Since Q is a deterministic function of X , $H(X, Q) = H(X)$. Apply the Shannon chain rule. $\square$

Corollary 9.2 (Boltzmann entropy in quotient fibers). *If $X \mid Q = q$ is uniform on a fiber of multiplicity $\Omega(q)$, then*

$$H(X \mid Q = q) = \log \Omega(q) \quad (23)$$

and

$$S_G = k_B H(Q) + \mathcal{E}_Q[k_B \log \Omega(Q)]. \quad (24)$$

Corollary 9.3 (Equality boundary). *The quotient Shannon entropy equals total Gibbs entropy in units of k_B if and only if*

$$H(X \mid Q) = 0, \quad (25)$$

equivalently, π is injective on $\text{supp } p$.

Corollary 9.4 (Resolution redistribution). *If Q_f refines Q_c , then*

$$H(Q_f) \geq H(Q_c), \quad H(X \mid Q_f) \leq H(X \mid Q_c), \quad (26)$$

while $H(X)$ remains fixed. Increased resolution transfers entropy from unresolved fibers to observable classes; it does not create total entropy.

10 Finite fixed-protocol attack

10.1 Design

A fixed six-mode protocol was declared in a G -orthonormal information eigenbasis:

$$\lambda = (0.18, 0.42, 0.91, 1.37, 2.15, 3.80), \quad \tau = 1. \quad (27)$$

The nuisance, stability, conditioning, coarse-graining, and transport masks were frozen before enumeration. The canonical resolved and admissible masks were

$$P_\tau : (0, 0, 0, 1, 1, 1), \quad (28)$$

$$W_{\text{obs|adm}} : (0, 0, 0, 1, 0, 0). \quad (29)$$

Every binary hard spectral projector mask and every binary final-sector mask were paired, giving

$$2^6 \times 2^6 = 4096 \quad (30)$$

competitors.

10.2 Elimination ledger

Table 2: Survivors in the exhaustive fixed-protocol attack.

Constraint	Survivors
Initial hard spectral/sector pairs	4096
Fixed-threshold consistency	64
Gate soundness	2
Gate completeness/maximality	1
Transport naturality	1
Product composition	1

Exactly one pair survived, equal to the canonical construction. Soundness alone left two competitors: the canonical one-dimensional sector and its zero strict subspace. Completeness removed the zero sector. Thus the finite attack did not merely fail to find alternatives; it isolated an explicit independence witness for the new axiom.

A second attack generated 128 random oblique similarities of the canonical projection. None simultaneously satisfied idempotence, metric self-adjointness, and commutation with the information operator.

Proposition 10.1 (Finite battery result). *Within the enumerated six-mode hard-spectral class, the fixed protocol admits exactly one observable-sector assignment satisfying threshold consistency, gate soundness and completeness, natural transport, and product composition.*

Proof. Exhaustive enumeration checks all 4096 binary spectral/projected-sector pairs. The survivor ledger above is generated deterministically by the accompanying script. The unique surviving masks coincide with the analytic construction. $\square$

Remark 10.2. *The computation is not a proof of the representation theorem. The theorem follows from the modular arguments above. The battery instead checks implementation, exhibits the necessity of completeness in the finite model, and searches the full stated binary competitor class for overlooked alternatives.*

11 Axiom roles and independence

Proposition 11.1 (Soundness does not imply completeness). *There exist fixed protocols for which more than one sound sector exists.*

Proof. If the canonical maximal sector is nonzero, both it and the zero subspace satisfy containment in every gate. Only the maximal sector satisfies completeness. $\square$

Proposition 11.2 (Completeness does not select a protocol). *Admissibility completeness is conditional on Γ and does not derive G , τ , N , S , C , or T from $\mathcal{E}$.*

Proof. Different protocols define different admissible families $\text{Adm}_\Gamma(R_\tau)$ and therefore may have different greatest elements. Completeness selects the greatest object within a declared family; it does not select the family. $\square$

Proposition 11.3 (Threshold consistency is independent of hard projection). *Idempotence, metric self-adjointness, and commutation with L admit every orthogonal spectral subprojection. Threshold consistency is required to select the specified upper spectral set.*

Proof. In a spectral basis, any binary choice on eigenspaces defines a commuting orthogonal projector. Only the threshold mask fixes which eigenspaces are retained. $\square$

12 Coherence, operator duration, and conditional proper time

12.1 Canonical duration on an observable sector

Definition 12.1 (Observable coherence semigroup). *Let $W_{\text{obs|adm}}$ be a finite-dimensional Hilbert sector. A basic coherence process is a strongly continuous family of self-adjoint contractions satisfying*

$$C_0 = I, \quad C_{t+s} = C_t C_s, \quad 0 \prec C_t \preceq I.$$

Exact zero modes are handled on the support or by extended duration $+\infty$.

Theorem 12.2 (Operator-duration characterization). *There is a unique $K \succeq 0$ such that $C_t = e^{-tK}$, and*

$$\boxed{\mathcal{T}_t = -\log C_t = tK}, \quad \mathcal{T}_{t+s} = \mathcal{T}_t + \mathcal{T}_s.$$

Under an admissible similarity $C'_t = UC_tU^{-1}$, functional calculus gives $\mathcal{T}'_t = U\mathcal{T}_tU^{-1}$.

Proof. Finite-dimensional semigroup theory gives $C_t = e^{tA}$. Self-adjoint contractivity gives $A = -K$ with $K \succeq 0$. Strict positivity permits the principal logarithm, and functional calculus is similarity-equivariant on the stated domain. $\square$

Theorem 12.3 (Extensive scalar coherence character). *Let $\chi_n : \text{PD}_n(\mathbb{C}) \rightarrow \mathbb{R}_{>0}$ be continuous, unitarily invariant, multiplicative on commuting positive operators, and extensive under direct sums, with $\chi_n(I) = 1$. Then*

$$\chi_n(A) = (\det A)^a$$

for one constant a independent of n . Strict clock monotonicity fixes the sign convention and $a > 0$. Hence

$$\tau_{\det}(t) = -\log \chi(C_t) = a \text{Tr } \mathcal{T}_t.$$

Proof. Unitary invariance reduces $\log \chi_n$ to a symmetric spectral function. Direct-sum extensivity makes it a sum of one-variable terms. Scalar multiplicativity gives the continuous Cauchy equation $g(xy) = g(x) + g(y)$, so $g(x) = a \log x$. $\square$

Non-claim 12.4 (Scalarization remains semantic). *The determinant clock is unique only in the extensive-character class. Modal expectations, operator norms, spectral extrema, and normalized traces answer different operational questions and require their own declarations.*

12.2 Noncommutative composition and memory

For ordered positive processes define $\mathcal{T}_{02} = -\log(e^{-\mathcal{T}_{12}}e^{-\mathcal{T}_{01}})$ where a consistent logarithm exists. In a convergent BCH neighbourhood,

$$\mathcal{T}_{02} - \mathcal{T}_{01} - \mathcal{T}_{12} = \frac{1}{2}[\mathcal{T}_{01}, \mathcal{T}_{12}] + O(\|\mathcal{T}\|^3).$$

This is an algebraic noncommutativity defect. It is neither necessary nor sufficient for operational non-Markovianity, which must be defined through the declared multitime statistics, process tensor, or divisibility criterion.

Theorem 12.5 (Universal affine clock). *A connected family of nondegenerate clock charts whose overlap functions are positive affine maps and satisfy identity and cocycle laws glues to a one-dimensional affine clock object, unique up to one global positive affine transformation. For elapsed durations only the positive scale remains.*

Proof. Fix one chart as gauge and propagate its coordinate through the overlap graph. Cocycle closure gives path independence. Any second compatible construction differs on the reference chart, and hence globally, by one positive affine automorphism. $\square$

Theorem 12.6 (Conditional proper-time realization). *Suppose independent clock mechanisms form the universal affine clock; held-out path comparisons are stable; directed causal order is independently observed; interval designs have sufficient rank; and one nondegenerate Lorentzian tensor field explains all tested clock-path integrals. Within that declared model class, the universal clock is identified up to global unit and origin with proper time on the tested timelike paths.*

Proof. Affine factorization fixes the common duration object. Full-rank interval data identify the local tensor within the competitor class, while causal data select orientation. Agreement on held-out path integrals identifies the calibrated duration functional with Lorentzian proper time up to the remaining affine gauge. $\square$

Non-claim 12.7 (No coherence–proper-time identity). *The theorem is conditional and does not show that arbitrary coherence loss equals relativistic proper time. Coherence duration, universal clock factorization, causal orientation, metric reconstruction, and physical identification are distinct layers.*

Theorem 12.8 (Hidden-extension invariance). *If a latent model is enlarged by an auxiliary factor invisible to every declared observation law, the enlarged and original systems have isomorphic observational quotients and identical natural RG structures. Observable Lorentzian rank therefore does not identify latent dimension.*

Proof. All auxiliary fibres lie inside equality-of-law classes and are collapsed by the quotient. Naturality transfers every downstream experiment-internal structure. $\square$

13 Unified representation theorem

Theorem 13.1 (Modular representation of classical, quantum, and temporal RG). *Let $(\mathcal{E}, \Gamma)$ be either a broad classical experiment or a finite-dimensional quantum experiment with a declared tester family. Then:*

1. *the equality-of-declared-law quotient and its factorization are unconditional;*
2. *Blackwell, quantum-channel, or deficiency comparison is available when the corresponding randomization category and topology are declared;*
3. *Fisher–Rao geometry is characterized only on its regular classical module;*

4. *quantum local geometry is a declared Petz monotone member or family, not a unique metric;*
5. *hard resolution is uniquely spectral once (F, G, τ) or its quantum analogue is fixed;*
6. *the final sector is unique when soundness and completeness select a greatest admissible subobject;*
7. *natural transports glue exactly when identity and cocycle laws hold;*
8. *a positive coherence semigroup adds canonical operator duration, while a scalar or physical clock requires the additional hypotheses stated above.*

Consequently every fully declared regular fibre has a unique RG realization up to protocol-preserving natural isomorphism; outside regular fibres, only the modules whose hypotheses hold are asserted.

Proof. Apply the general quotient theorem, the relevant comparison module, the applicable classical or quantum metric characterization, spectral functional calculus, greatest-subobject uniqueness, and cocycle gluing. The time claims follow from semigroup generation, logarithmic functional calculus, the determinant-character theorem, and affine-clock gluing. No implication is used outside its stated domain. $\square$

14 Tautological closure and exact locus of choice

Theorem 14.1 (Fibrewise closure theorem). *Within the regular finite classical domain and for a fixed bundled input $(\mathcal{E}, \Gamma)$ satisfying the stated existence hypotheses, every component of $\text{RG}(\mathcal{E}, \Gamma)$ is either (i) forced by an established universal, characterization, spectral, greatest-subobject, or gluing property, or (ii) explicitly present in Γ . No third, undeclared choice remains.*

Proof. The quotient is forced by equality-of-law extensionality and its universal property. The metric is forced, up to declared normalization, by the imported classical metric characterization. The hard projector is forced by spectral functional calculus and threshold consistency. The admissible sector is forced by soundness plus completeness as the greatest common admissible subobject. Transported copies and overlap identifications are forced relative to the declared morphisms by naturality and cocycle closure. All remaining choices—comparison scale, threshold, nuisance convention, gate class, and transition data—are components of Γ and are therefore visible assumptions rather than hidden degrees of freedom. $\square$

Corollary 14.2 (Exact tautology statement). *RG is mathematically tautological only in the disciplined sense that its output is the uniquely structured consequence of its declared input and axioms. It is not vacuous: changing or violating an input axiom changes, destroys, or renders non-unique the resulting object.*

Non-claim 14.3 (No derivation of the contract). *The closure theorem does not derive the observation system or protocol from nothing. It proves absence of undeclared freedom after they are specified.*

15 Interpretation

The representation theorem identifies RG with neither a bare quotient nor a universal final sector. The quotient is universal at the domain layer. The final sector is maximal only inside a fixed protocol fibre. The complete logical order is

$$(\mathcal{E}, \Gamma) \longrightarrow Q_{\mathcal{E}} \longrightarrow g_F \longrightarrow P_{\tau} \longrightarrow W_{\text{obs|adm}} \longrightarrow \{T_{ij}\}. \quad (31)$$

The word “canonical” must therefore be read fibrewise. A protocol is to RG what a metric is to Riemannian geometry: declared structure whose consequences are characterized. This is weaker than protocol-free uniqueness and stronger than presenting an arbitrary construction. It says that no additional undeclared choice remains after the complete admissibility contract and the RG axioms are fixed.

The theorem schema is modular. A future quantum version could replace the classical Fisher module with a characterized quantum distinguishability module while retaining quotient extensionality, hard or graded spectral resolution, admissibility maximality, and natural transport, provided the appropriate categories and subobject meets exist. No such extension is proved here.

16 Non-claims and falsification criteria

Non-claim 16.1 (No protocol-free uniqueness). *The paper does not assert that $\mathcal{E}$ determines Γ , or that all observation geometries are RG.*

Non-claim 16.2 (No universality of $W_{\text{obs|adm}}$). *The universal object is the equality-of-law quotient for experiment-internal maps. The final admissible sector is a maximal subobject relative to a fixed protocol, not an initial or terminal object across all protocols.*

Non-claim 16.3 (No unrestricted Fisher theorem). *Fisher–Rao is imported only in the regular finite classical domain under the stated invariance and normalization assumptions. Singular, infinite-dimensional, non-dominated, and quantum experiments require separate metric modules.*

Non-claim 16.4 (No automatic global uniqueness). *Atlas uniqueness is relative to declared transition data satisfying identity and cocycle laws. Different undeclared connections or holonomy classes are different protocols, not rival realizations in one fibre.*

Non-claim 16.5 (No physical ontology). *The quotient expresses operational indistinguishability. It neither denies latent representatives nor identifies the observable geometry with spacetime, quantum state reality, or any physical ontology.*

Non-claim 16.6 (No infinite-dimensional maximality without hypotheses). *In general categories or topological vector spaces, arbitrary gate intersections may fail to belong to the intended regularity class or may require closure. The finite linear greatest-subobject proof does not automatically extend.*

Falsification and revision conditions. The stated characterization fails or requires revision if:

1. an experiment-internal quantity varies inside an equality-of-law class;
2. a metric satisfying the stated classical module is not proportional to Fisher–Rao;
3. two distinct hard projectors satisfy all fixed (F, G, τ) selector axioms off the spectrum;
4. two distinct greatest admissible subobjects exist in the same fixed gate family;
5. covariant transport changes the characterized object beyond natural isomorphism;
6. direct and composite overlap transports disagree;
7. a claimed stable sector has no positive margin from the declared discriminant.

17 Open problems

1. Establish closed or completed greatest admissible subobjects for non-dominated, infinite-dimensional, nonlinear, and sheaf-valued experiment categories.
2. Prove quantitative continuity of selected RG outputs under Le Cam deficiency and determine sharp moduli near discriminants.
3. Characterize metric and spectral modules for quantum channels rather than state families, including ancilla stability.
4. Extend the quotient and admissibility construction to testers of instruments, quantum combs, and process tensors, with a genuine operational memory obstruction.
5. Determine which explicit decision, estimation, asymmetry, thermodynamic, or resource axioms select a Petz metric member.
6. Validate or falsify coherence-derived clocks against independent mechanisms and held-out relativistic paths.

18 Legacy finite-fibre open problems

1. Prove existence of greatest admissible subobjects for nonlinear, stratified, and sheaf-valued protocol categories.
2. Determine the correct closure operation for infinite-dimensional gate intersections and establish perturbation-stable maximality.
3. Extend the representation theorem to singular and rank-changing experiments.
4. Formulate a quantum experiment category and replace the classical metric module by an explicitly characterized monotone quantum module.
5. Separate protocol-preserving natural equivalence from equivalence under protocol refinement or Blackwell comparison.
6. Determine when decision, resource, or physical axioms select a subfamily of protocols; this is external to the present fibrewise theorem.

19 Conclusion

The correct classical identification is now precise. Resolution Geometry is the canonical observable geometry of a declared admissible experiment, not the unique protocol-free geometry of observation. Quotient universality fixes the observable domain. The classical metric characterization fixes Fisher–Rao distinguishability in its domain. Spectral theory fixes the hard resolved projector. The newly explicit completeness axiom fixes the final sector as the greatest resolved subobject satisfying every declared gate. Naturality and cocycle laws fix transport and gluing relative to the contract.

Thus the defensible uniqueness statement is fibrewise:

$$(\mathcal{E}, \Gamma) \implies \text{RG}(\mathcal{E}, \Gamma) \quad \text{up to protocol-preserving natural isomorphism.} \quad (32)$$

The finite battery supports this statement, exposes completeness as necessary, and finds no competing hard-spectral realization in the exhaustive class. Protocol selection remains outside the theorem by design.

Reproducibility

The deterministic supporting artifacts are:

- `rg_fixed_protocol_uniqueness_attack.py`;
- `rg_fixed_protocol_uniqueness_attack.json`.

The seed is 20260805. The exhaustive binary enumeration itself is seed-independent; the seed controls the supplementary oblique-projector attack.

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A Battery pseudocode

```

fix information spectrum, threshold, and all protocol gates
canonical_P = spectral_mask(information > threshold)
canonical_W = canonical_P AND all_gate_masks

for P in all binary spectral projectors:
    for W in all binary sector masks:
        require P == canonical_P                # threshold consistency
        require W subset_of (P AND all_gates)    # soundness
        require W == (P AND all_gates)          # completeness
        require covariance under transported protocol
        require direct-sum composition
        retain(P,W)

assert exactly one retained pair
assert retained pair == (canonical_P, canonical_W)

```

B Machine-readable result summary

all competitors:	4096
survive threshold consistency:	64
survive gate soundness:	2
survive gate completeness:	1
survive natural transport:	1
survive product composition:	1
random oblique projectors tested:	128
oblique survivors of all projector rules:	0